READ THIS FIRST

One complete, traceable run

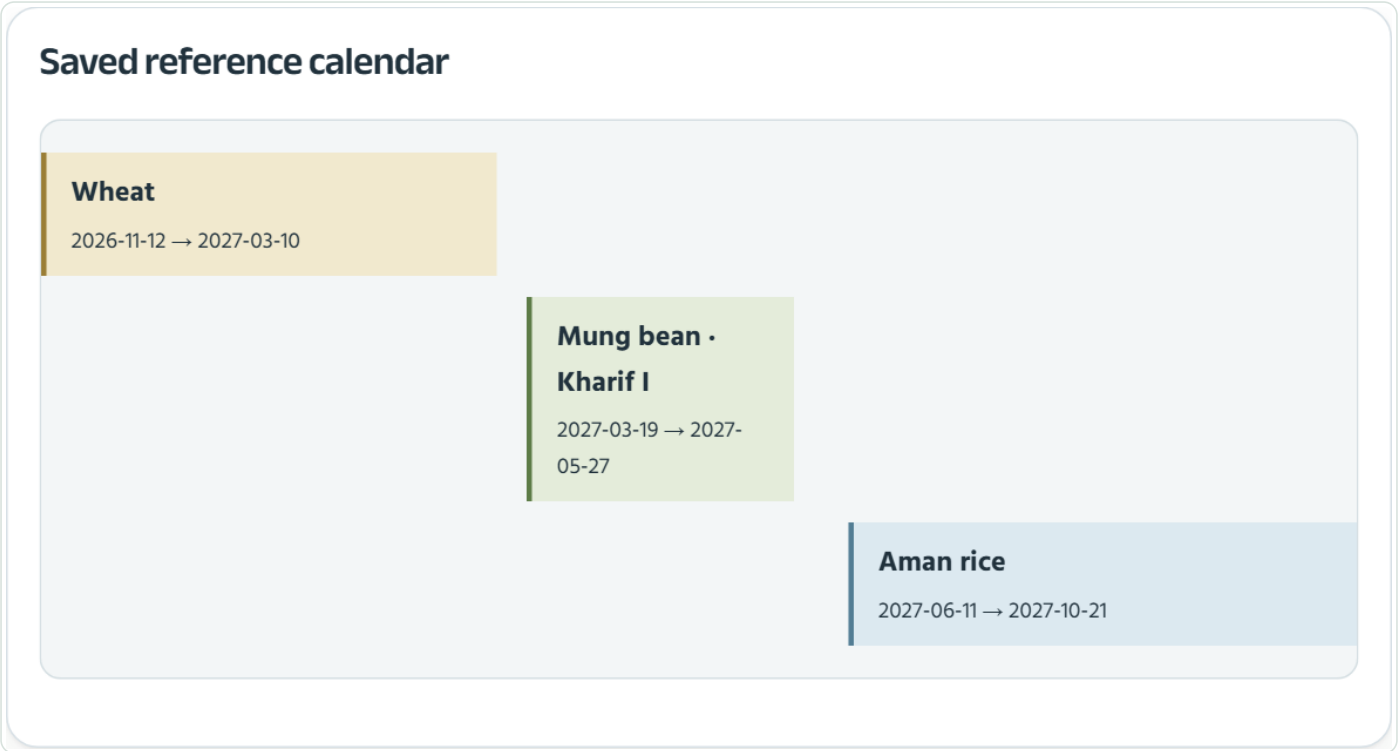
This report follows one input sequence from a blank profile through crop selection, a saved reference calendar, tracking and an exported record. Every screenshot comes from the running RITU website.

Definitive output: Wheat > Mung bean - Kharif I > Aman rice. The exact dates and durations are transcribed later from the saved output.

Tested: 216 completed browser-entry scenarios across desktop and phone layouts. Separate evidence below distinguishes captured real-provider replay used for repeated QA from normal live-provider calls used for this walkthrough.

What this establishes: the tested interface, constraints, calculations and persistence behave as reported. It does not establish the best crop, optimum sowing date, yield, irrigation dose or agronomic effectiveness.

Case identity: a published Rajshahi research-site example. Historical plot facts are combined with a deliberately selected planning date and current regional environmental context. This is not an independently surveyed current farm.



The saved calendar is a stored reference plan. Future reference dates are distinct from actual completed activities.

INPUT AND PROVENANCE LEDGER

Exactly what was entered

The research source is Chaki et al., Field Crops Research 275 (2022), 108344, printed page 3. Its Rajshahi trials were conducted in 2016-2017. Published coordinates are approximate site coordinates; the 30 m2 value describes a subplot.

Location, area and the historical rainfed treatment are source-grounded. Soil/drainage remain unknown. The planning date, preferences and crop choices are walkthrough decisions, not observations or NASA recommendations.

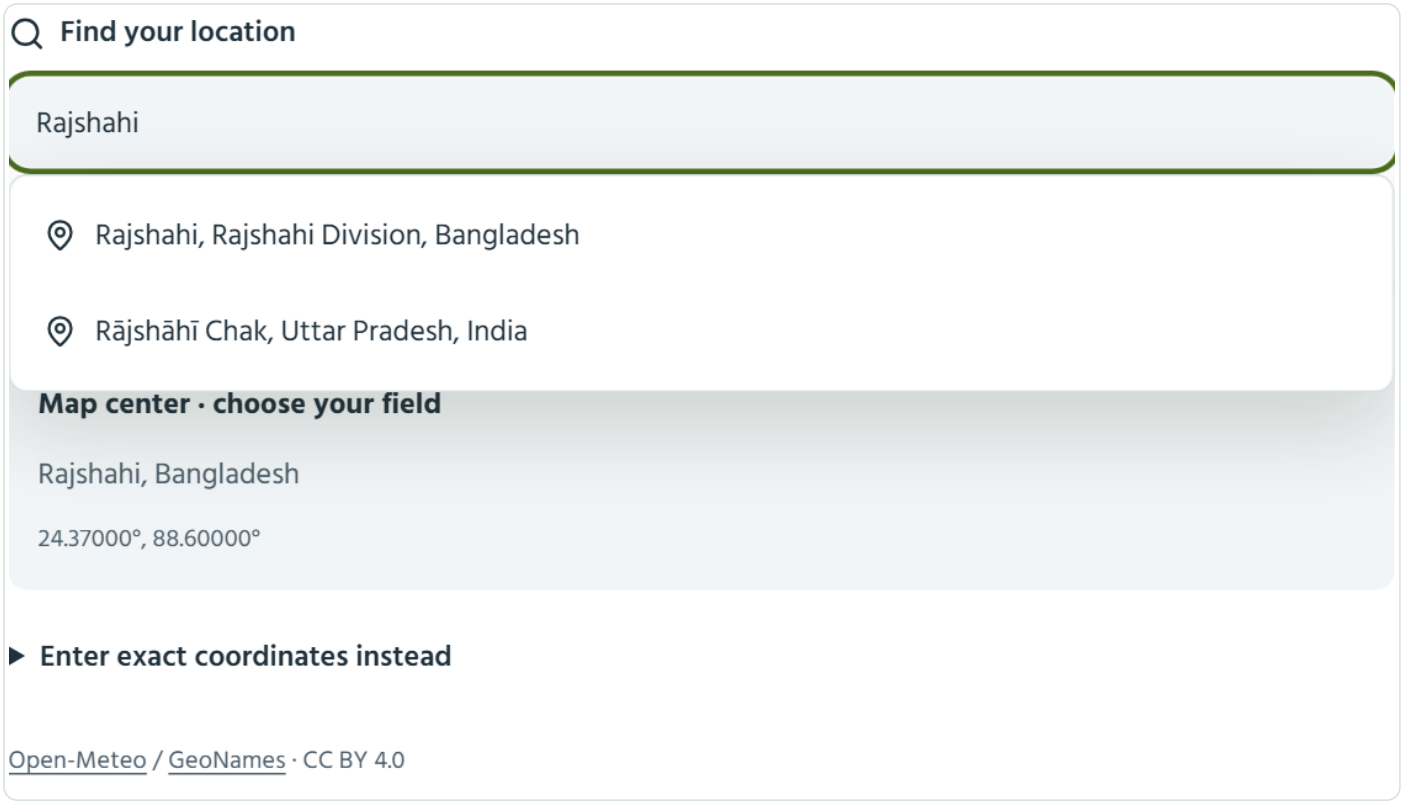
Input	Entered value / evidence status
Site	24.36667 N, 88.65000 E - historical published research site
Field area	30 square metres / 0.003 hectares - historical subplot
Water	Rainfed - historical CS6 category, current supply unverified
Soil / drainage	Not sure / Not sure
Previous crops	Wheat; mung bean - historical familiarity only
Plan from	2026-10-31 - chosen planning start
Priority	More botanical families - chosen preference
Crop choices	Wheat; mung bean; Aman rice
Household / labor constraints	None entered; availability not independently established
Actual activity dates	Blank - no current planting or harvest verified

STEP 1 OF 4 / LOCATION SEARCH

Find the correct place

Typing Rajshahi queries the live place-search service. Select the correct country and region before positioning a farm pin. A town result does not locate a field boundary.

For this run, the published research-site coordinates are then entered explicitly in the next screen. This makes the location choice reproducible.



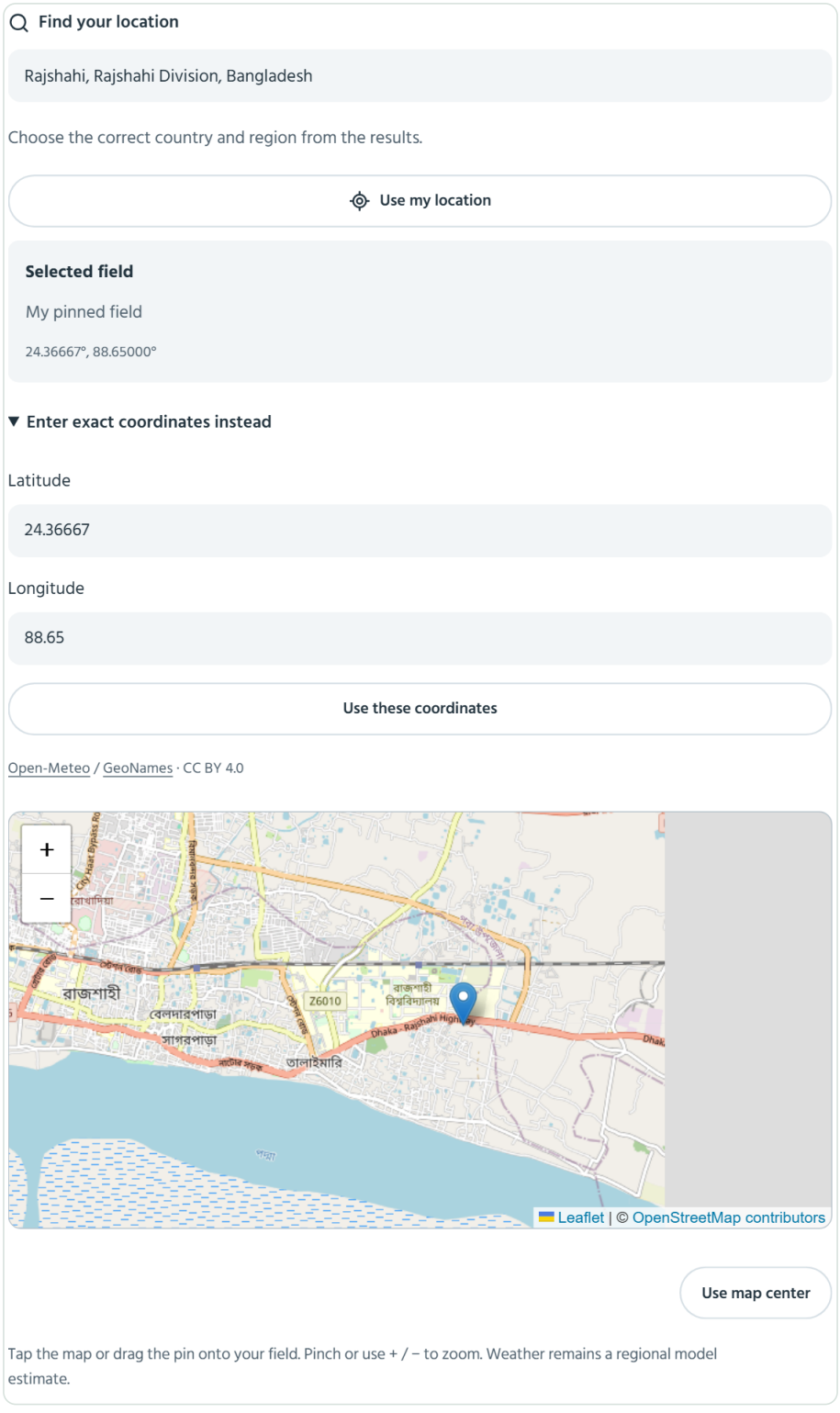
Typed Rajshahi into the live location search. Suggestions are place matches, not farm boundaries.

STEP 1 OF 4 / COORDINATES

Confirm the research-site pin

Latitude 24.36667 N and longitude 88.65000 E are the decimal conversion of the source site coordinates. Displaying five decimals does not increase the original source accuracy.

Map location, NASA grid location and the forecast grid are different concepts. RITU uses the selected point to request regional environmental data; it does not measure the plot.



The pin uses the approximate published research-site coordinates. The map is a location reference, not a parcel survey.

STEP 1 OF 4 / FIELD DETAILS

Enter area, district and planning date

The area unit is Square metres and the entered area is 30. The app converts this to 0.003 ha. This is an experimental subplot, not a 30-hectare commercial field.

Rajshahi selects the supported regional reference set. Plan from 2026-10-31 is the chosen starting point for looking ahead; it is not evidence that sowing happened that day.



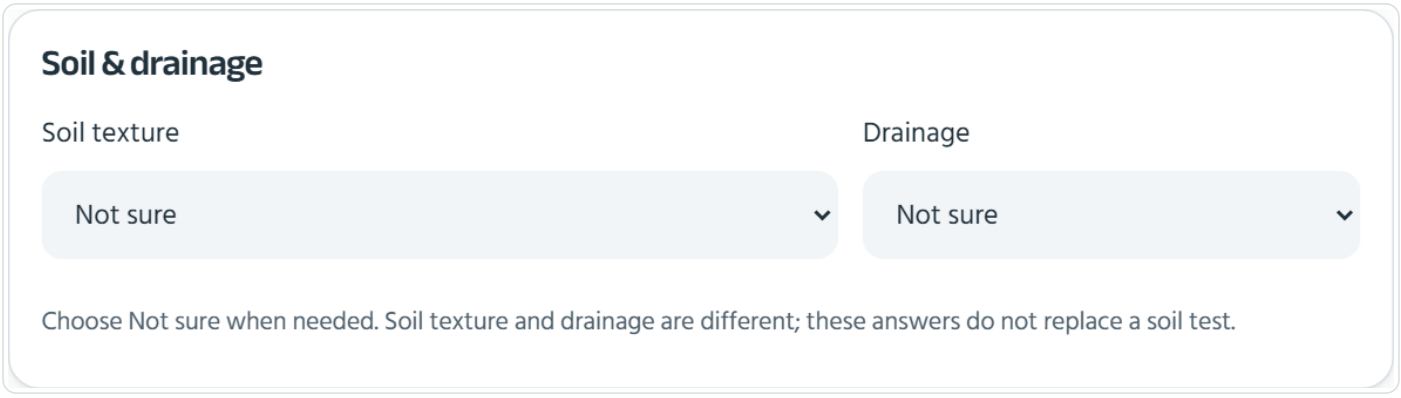
A 30 square metre research subplot is 0.003 hectares. The selected planning date is a walkthrough choice, not a recorded planting date.

STEP 1 OF 4 / SOIL AND WATER

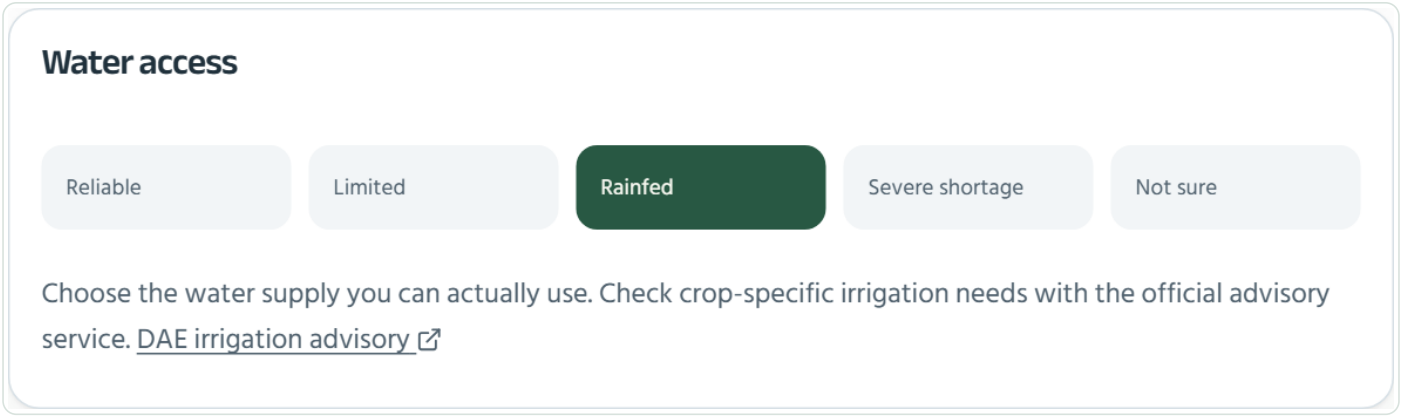
Keep unknowns visible

Soil texture and drainage remain Not sure. The source contains more detailed historical soil properties, but those have not been treated as a current field test or forced into a coarse label.

Rainfed represents the documented historical CS6 treatment. Present-day irrigation access has not been independently verified. Selecting this category does not establish that a future rice crop will succeed.



Soil texture and drainage stay Not sure: no current field measurement has been obtained.



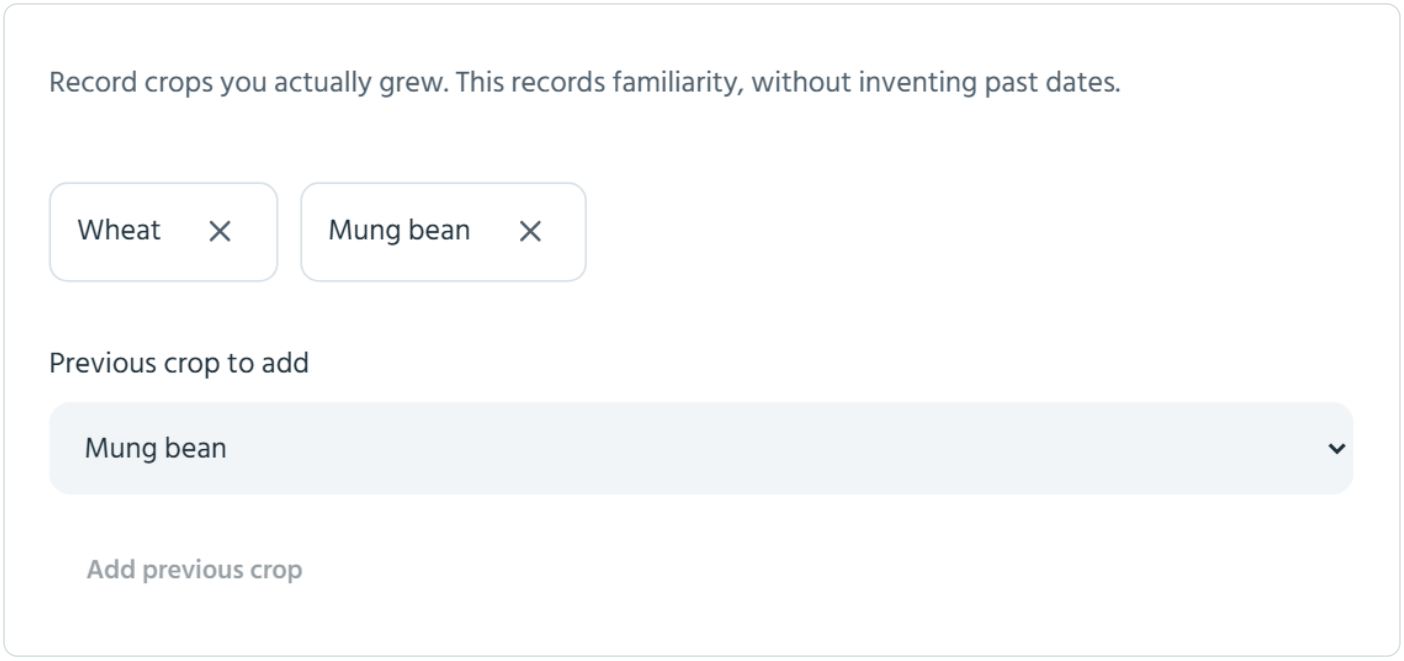
Rainfed reproduces the historical CS6 treatment category. It does not verify the water supply at this site today.

STEP 1 OF 4 / OPTIONAL INPUTS

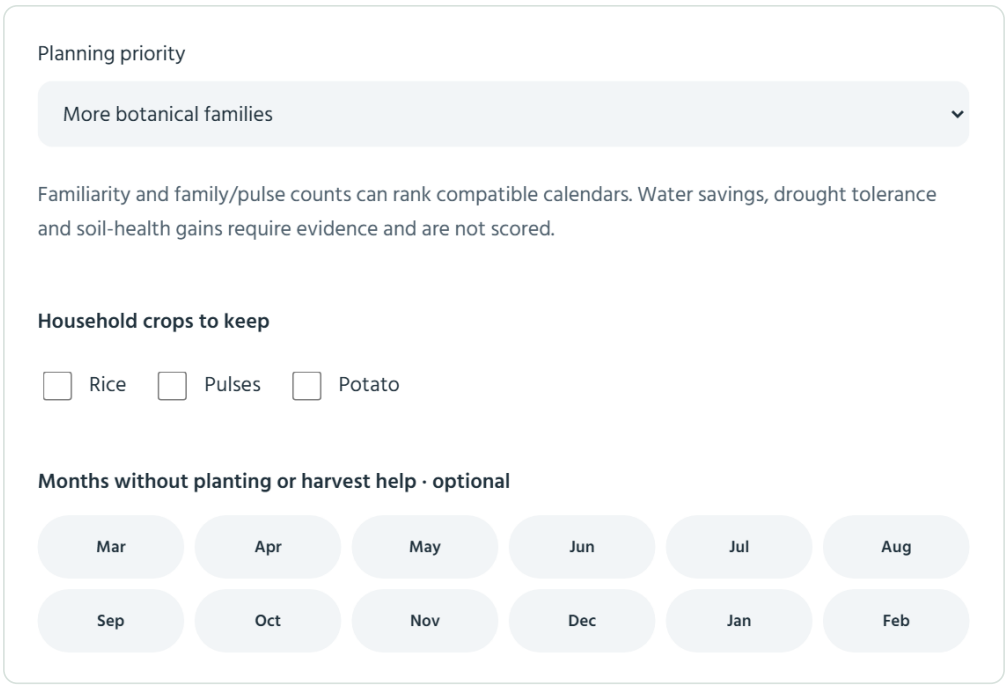
Record familiarity and preferences

Previous crops record familiarity without creating invented past dates. More botanical families is the chosen ranking preference.

No household crop requirement or unavailable-help month was entered. This means no such constraint was supplied, rather than proof that labor is always available. These choices can change which compatible calendars remain.



Wheat and mung bean are documented in the historical site description. This records familiarity, without inventing past planting dates.



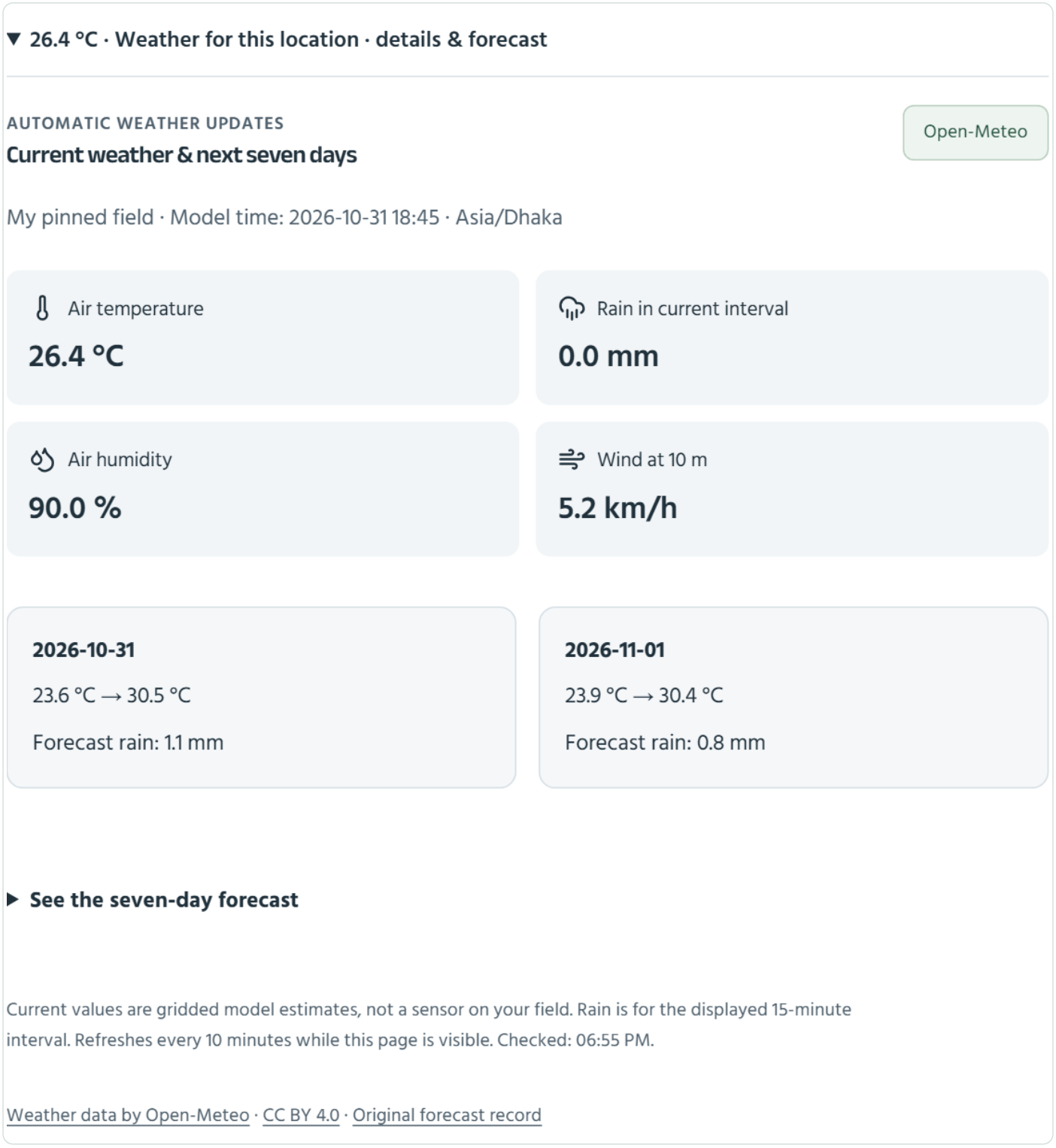
More botanical families is the chosen comparison preference. No household or labor restrictions are asserted; empty checkboxes mean no such constraints were entered.

STEP 1 OF 4 / CURRENT MODEL ESTIMATES

Read current weather correctly

The figures here are Open-Meteo model estimates for the selected location. They are separate from NASA history. The displayed model time, interval and timezone identify when the values apply.

Air temperature is in degrees Celsius, air humidity in percent, interval rainfall in millimetres and wind at 10 metres in km/h. Forecast rain is not measured root-zone moisture or an irrigation instruction.



Current model estimates and seven-day forecasts come from Open-Meteo. Forecast rainfall is not measured soil water.

STEP 1 OF 4 / FORECAST

Explore the seven-day forecast

Each row is a forecast day. Temperature is a daily minimum-to-maximum range. Rain is expected daily accumulation; rain probability is the likelihood reported by the provider.

Forecasts can change. Recheck the source time before using the information. RITU does not turn this table into a precise watering or treatment prescription.

▼ **See the seven-day forecast**

Forecast for the farm's local dates. Amounts and probabilities are estimates, not guarantees.

Date	Low → high	Rain	Rain chance
Sat, Oct 31	23.6 °C → 30.5 °C	1.1 mm	87.0 %
Sun, Nov 1	23.9 °C → 30.4 °C	0.8 mm	51.0 %
Mon, Nov 2	23.0 °C → 30.6 °C	3.6 mm	33.0 %
Tue, Nov 3	22.4 °C → 30.6 °C	0.0 mm	0.0 %
Wed, Nov 4	23.3 °C → 28.5 °C	0.0 mm	2.0 %
Thu, Nov 5	21.7 °C → 26.1 °C	0.0 mm	2.0 %
Fri, Nov 6	19.9 °C → 29.0 °C	0.0 mm	4.0 %

Seven local forecast dates: daily temperature range, expected rain accumulation and rain probability. These values can change.

STEP 1 OF 4 / NASA SOURCE YEAR

Separate history from forecasts

NASA POWER 2025 is a complete historical year at the requested regional grid point. The original-record link exposes the actual provider request.

The historical year does not replace the planning date. NASA meteorology is gridded model/reanalysis information, not a sensor on this subplot, a soil-texture test or a record of its irrigation supply.

▼ **NASA seasonal context & sources**

NASA POWER historical year 2025 · 24.37°, 88.65°. Original NASA record

NASA history provides seasonal context and has a publication delay. Current weather and forecasts above come separately from Open-Meteo. Neither source measures your field's soil texture or irrigation supply.

NASA POWER 2025 is historical regional context. Its source year is separate from the future planning date.

STEP 2 OF 4 / PUBLISHED CROP REFERENCE

Choose Wheat

This crop was explicitly selected in its reference season. The app reserves 119 days, from 2026-11-12 through 2027-03-10 inclusive.

These timings come from the linked DAE/BAMIS regional crop-weather reference. They are not NASA-generated optimum dates. For transplanted rice, nursery and main-field arrangements need separate local confirmation.

The photograph is a species/reference image. It is not a satellite detection of this plot or a prediction of what its crop will look like.



Photo: Patrick Alexander from Las Cruces, NM · CC0 · Source

☑ **Wheat**

119 days in the regional reference

Next computed reference interval: 2026-11-12 → 2027-03-10

▶ **Why offered, source & timing**

Wheat: a selected published regional reference, not a NASA crop-suitability verdict.

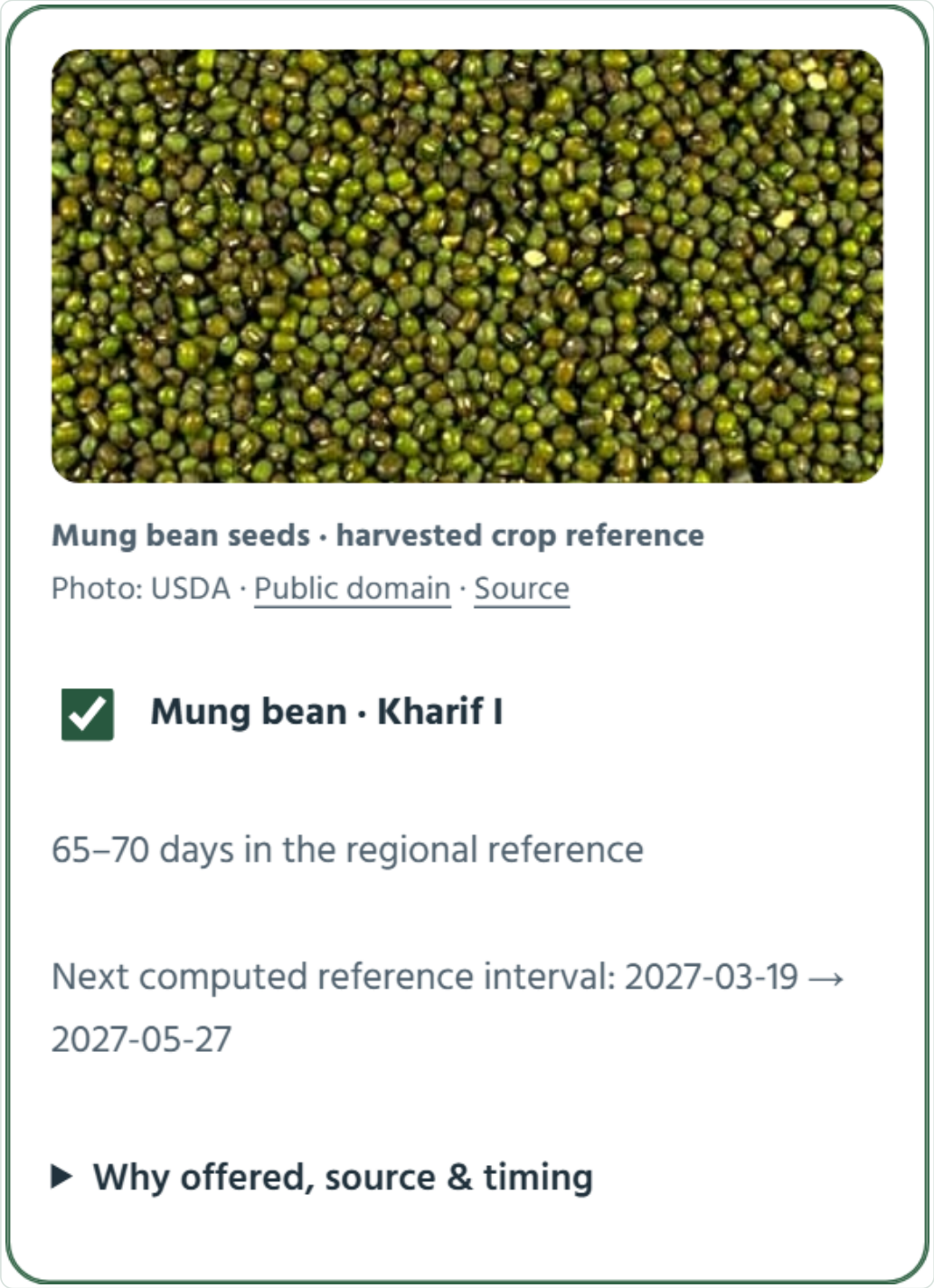
STEP 2 OF 4 / PUBLISHED CROP REFERENCE

Choose Mung bean - Kharif I

This crop was explicitly selected in its reference season. The app reserves 70 days, from 2027-03-19 through 2027-05-27 inclusive.

These timings come from the linked DAE/BAMIS regional crop-weather reference. They are not NASA-generated optimum dates. For transplanted rice, nursery and main-field arrangements need separate local confirmation.

The photograph is a species/reference image. It is not a satellite detection of this plot or a prediction of what its crop will look like.



Mung bean · Kharif I: a selected published regional reference, not a NASA crop-suitability verdict.

STEP 2 OF 4 / PUBLISHED CROP REFERENCE

Choose Aman rice

This crop was explicitly selected in its reference season. The app reserves 133 days, from 2027-06-11 through 2027-10-21 inclusive.

These timings come from the linked DAE/BAMIS regional crop-weather reference. They are not NASA-generated optimum dates. For transplanted rice, nursery and main-field arrangements need separate local confirmation.

The photograph is a species/reference image. It is not a satellite detection of this plot or a prediction of what its crop will look like.



Photo: Sajib5467 · CC0 · Source

☑ **Aman rice**

133 days in the regional reference

Next computed reference interval: 2027-06-11 → 2027-10-21

▶ **Why offered, source & timing**

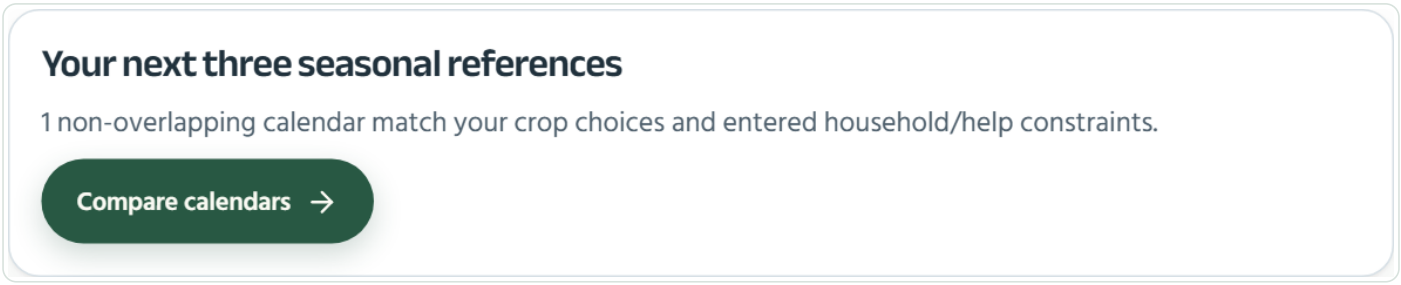
Aman rice: a selected published regional reference, not a NASA crop-suitability verdict.

STEP 2 OF 4 / COMPATIBILITY GATE

Check that the choices fit

The calendar action becomes available only when the selected crops can form a compatible source-based sequence under the entered constraints. A disabled action should be investigated rather than bypassed.

A different selection may require explicitly leaving a season unplanted. RITU preserves full durations instead of shortening a crop to force a result.



The chosen references must form a complete compatible sequence before the calendar link is enabled.

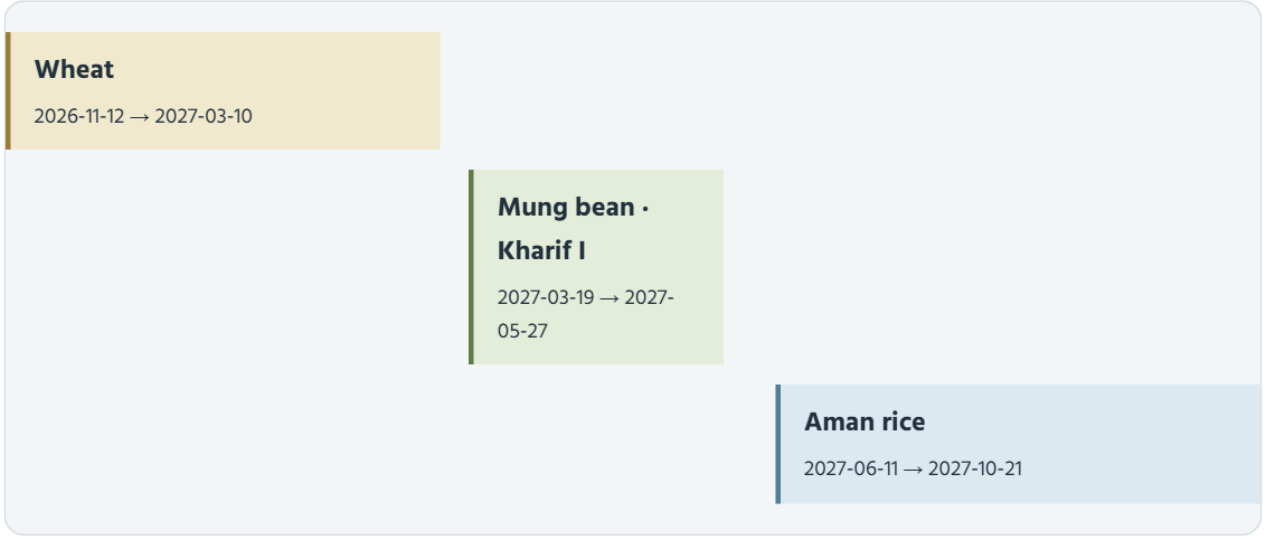
STEP 3 OF 4 / ACTUAL OUTPUT

Understand the definitive calendar

This is the saved output for the definitive inputs. The individual rows show inclusive display dates and full reserved durations. The timeline also shows unallocated gaps between crops.

Wheat and rice share a botanical family, so a three-crop sequence need not contain three families. Pulse inclusion and family counts describe the sequence; they do not quantify soil-health improvement.

Your planned reference sequence



1. **Wheat**

   2026-11-12 → 2027-03-10 · 119 days reserved

   Regional reference from sowing through harvest. Duration is 119 days; cultivar and actual maturity require confirmation.

2. **Mung bean · Kharif I**

   2027-03-19 → 2027-05-27 · 70 days reserved

   Regional calendar begins with germination. Confirm the sowing date and cultivar locally; no cultivar is named in this calendar.

3. **Aman rice**

   2027-06-11 → 2027-10-21 · 133 days reserved

   133 days includes the seedbed phase, followed by transplanting. Nursery dates are not field transplanting dates; confirm nursery arrangements and the variety.

Unallocated intervals between crops: 2027-03-11 → 2027-03-18 · 2027-05-28 → 2027-06-10. These are gaps, not a claim about soil cover or managed fallow.

The definitive output preserves the full published crop durations, orders the crops and explicitly displays the unallocated gaps.

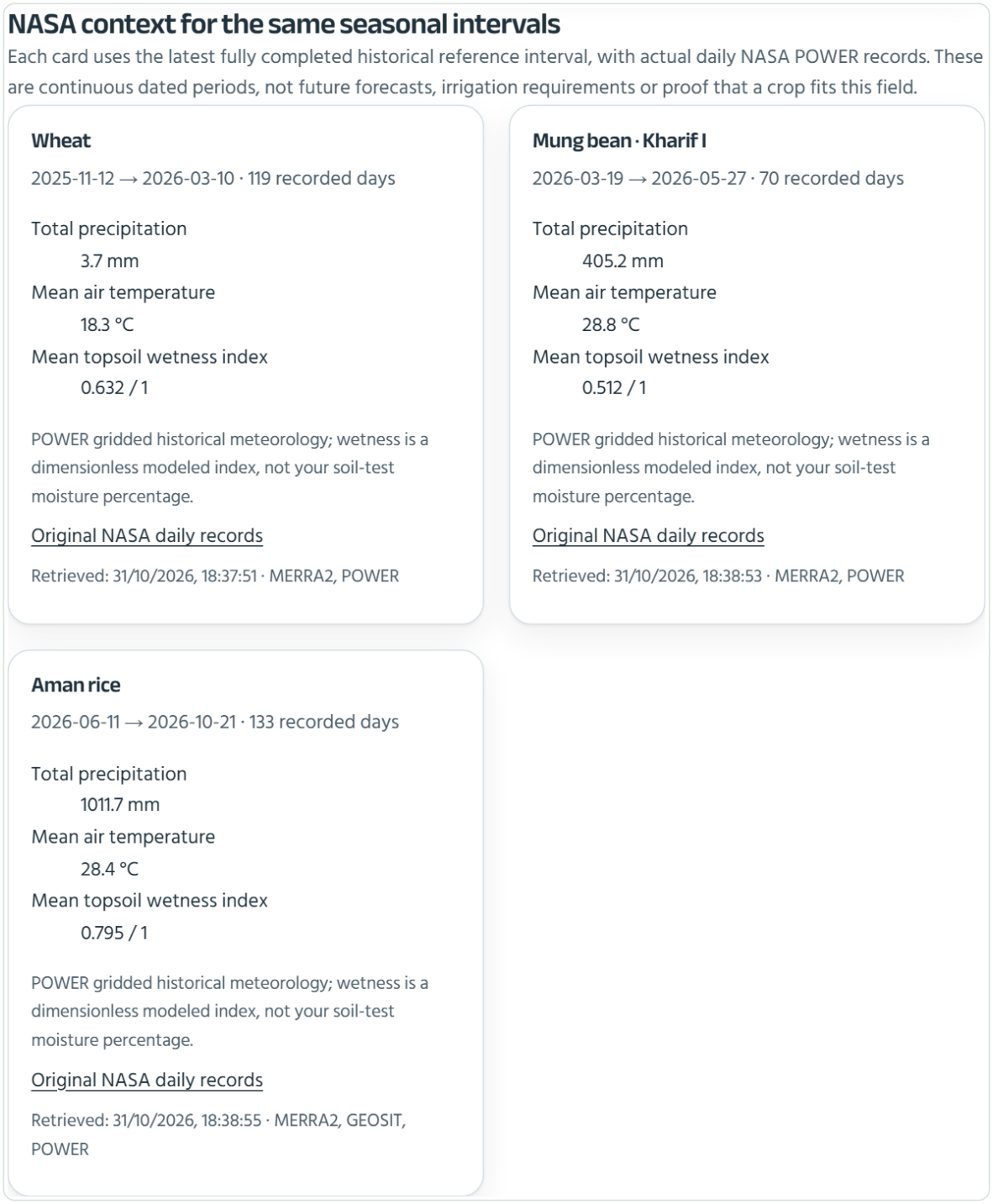
STEP 3 OF 4 / AUTHENTIC HISTORICAL RECORDS

Read the NASA interval cards

Each card describes the latest fully completed historical reference interval for that crop. Those historical dates can differ from the future calendar dates. The numbers below are retained from actual API responses.

Precipitation is the sum across the interval in millimetres. Temperature is the mean daily air temperature in degrees Celsius. Wetness is a dimensionless modeled topsoil index, not a field moisture percentage or SMAP retrieval.

Source request links, collection times and provider names make the values traceable. In the current app these values are contextual: they do not determine crop eligibility or rank the calendars.



NASA context for the same seasonal intervals

Each card uses the latest fully completed historical reference interval, with actual daily NASA POWER records. These are continuous dated periods, not future forecasts, irrigation requirements or proof that a crop fits this field.

Wheat

2025-11-12 → 2026-03-10 · 119 recorded days

Total precipitation
3.7 mm
Mean air temperature
18.3 °C
Mean topsoil wetness index
0.632 / 1

POWER gridded historical meteorology; wetness is a dimensionless modeled index, not your soil-test moisture percentage.

Original NASA daily records

Retrieved: 31/10/2026, 18:37:51 · MERRA2, POWER

Mung bean · Kharif I

2026-03-19 → 2026-05-27 · 70 recorded days

Total precipitation
405.2 mm
Mean air temperature
28.8 °C
Mean topsoil wetness index
0.512 / 1

POWER gridded historical meteorology; wetness is a dimensionless modeled index, not your soil-test moisture percentage.

Original NASA daily records

Retrieved: 31/10/2026, 18:38:53 · MERRA2, POWER

Aman rice

2026-06-11 → 2026-10-21 · 133 recorded days

Total precipitation
1011.7 mm
Mean air temperature
28.4 °C
Mean topsoil wetness index
0.795 / 1

POWER gridded historical meteorology; wetness is a dimensionless modeled index, not your soil-test moisture percentage.

Original NASA daily records

Retrieved: 31/10/2026, 18:38:55 · MERRA2, GEOSIT, POWER

These are the latest fully completed historical source intervals. They are not forecasts for the new calendar.

STEP 3 OF 4 / EXPLANATION

See why this calendar appears

The explanation connects published dates, compatibility, household/help constraints and simple counts. Current ranking supports familiarity, botanical-family count and pulse inclusion.

There is no validated water-saving, drought-resilience, yield or soil-health-gain score. The explanation should therefore be read as a compatibility account, not a scientifically proven recommendation.

▼ **Why this calendar?**

- Every interval comes from the linked official Rajshahi-region calendars. Full source durations are retained and do not overlap.
- Entered household requirements and unavailable-help months are checked. No silent removal of requirements.
- 2 botanical families. Pulse presence and familiarity are counts, not predictions of soil improvement or yield.
- Recorded soil: unknown; drainage: unknown; water access: rainfed. Field suitability, exact irrigation requirements and cultivar response remain unverified.

Wheat · DAE/BAMIS

Mung bean · Kharif I · DAE/BAMIS

Aman rice · DAE/BAMIS

Separate historical NASA context: NASA POWER 2025. This history is not a forecast or crop-suitability rule.

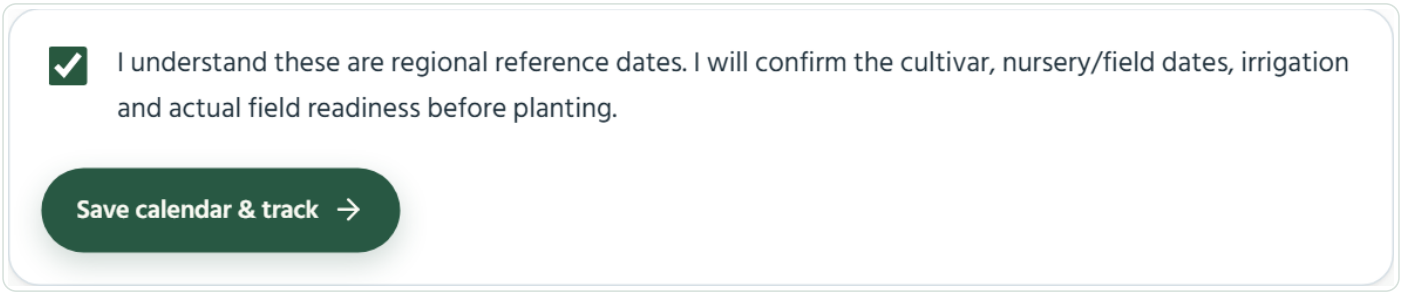
Reasons describe calendar compatibility, counts and the supplied constraints. NASA history is contextual and does not determine the ranking.

STEP 3 OF 4 / SAVED REFERENCE PLAN

Acknowledge and save

The acknowledgement records that you understand these are regional reference dates requiring cultivar, field-readiness, nursery and water checks. It does not certify those checks have already been performed.

Saving stores this plan and its site in this browser. It does not create a cloud account, survey the field or record a completed planting.



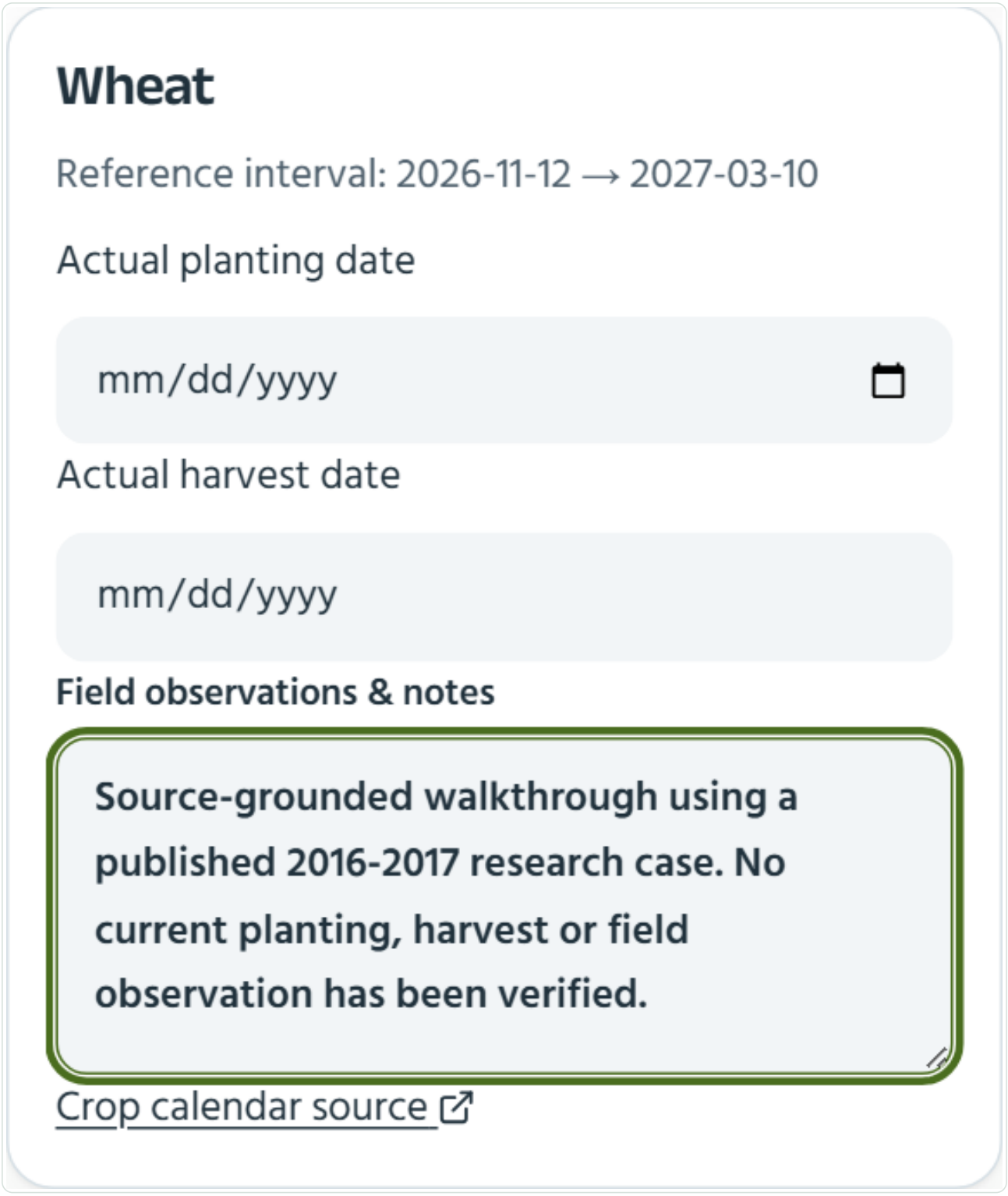
The confirmation acknowledges that these are regional reference dates requiring field and cultivar checks. Saving does not prove suitability.

STEP 4 OF 4 / FIELD RECORDS

Track facts without inventing activities

Actual planting and harvest dates remain blank because no current field activity was observed. A future reference date must not be entered as an already completed activity.

The administrative note identifies the historical source and absence of current observations. The note survived a reload. Export all records produced a JSON file that was opened and checked against the saved state.



Actual planting and harvest remain blank. The saved note documents the source and the absence of current field observations.

RESULT LEDGER

Read the final output as a whole

The following tables transcribe the saved reference plan and the authentic environmental responses, so the results remain readable without zooming screenshots.

Calendar end dates below are inclusive; the internal end boundary is exclusive. This distinction prevents double-counting a shared boundary day.

Crop	Start / inclusive finish / reserved days
Wheat	2026-11-12 to 2027-03-10 / 119 days
Mung bean - Kharif I	2027-03-19 to 2027-05-27 / 70 days
Aman rice	2027-06-11 to 2027-10-21 / 133 days

Botanical families: 2. **Pulse included:** yes. **Gaps:** 2027-03-11 to 2027-03-18; 2027-05-28 to 2027-06-10

Historical interval	Rain / mean temperature / topsoil wetness
Wheat 2025-11-12 to 2026-03-10	3.7 mm / 18.3 C / 0.632
Mung bean - Kharif I 2026-03-19 to 2026-05-27	405.2 mm / 28.8 C / 0.512
Aman rice 2026-06-11 to 2026-10-21	1011.7 mm / 28.4 C / 0.795

Weather model time: 2026-10-31T18:45 (Asia/Dhaka). **Air temperature:** 26.4 C. **Interval rain:** 0 mm. These are separate model estimates, not NASA observations.

QA / VERIFIED SCOPE

What the testing actually covered

The 216-case matrix comprises 108 distinct profiles on each of two viewports: four water settings, three drainage settings, three soil settings and three priorities. All input values were entered through browser controls.

Additional variation covers hectares / square metres / decimal conversion, January and leap-day starts, the upper supported planning year, selected previous crops, every offered calendar, source durations, non-overlap, family counts, blocked crop constraints, saving and reload persistence.

Exact freshly captured provider records were replayed for repeated matrix cases; their figures and timestamps were not changed. Community map tiles were suppressed in that matrix. Those scenarios are QA settings, not 216 measured farms.

The definitive screenshots use the normal running application with live provider requests and normal map tiles. No test route interception or prefilled farm state is used for that run.

A passing software test is not farmer usability research, an agronomic field trial or a guarantee that no other defect exists. Remaining limitations include regional timing, missing current field measurements and NASA history not driving ranking.

Check	Verified result
Matrix full run	216 passed; 0 unexpected failures
Definitive browser errors	0
Saved plan	3 intervals; no overlapping reserved dates
Environmental cards	All three selected crop intervals returned actual NASA records
Storage / export	Note retained after reload; JSON export verified

Separate domain/API run: Test Files 18 passed (18) / Tests 208 passed (208)

Initial test attempt: one historical combinatorial test exceeded its 30-second timeout while the browser matrix was running. The full serial rerun passed all 208 tests in 20.22 seconds. No timeout setting or app behavior was changed.

Separate existing acceptance suite: 68 passed; 0 unexpected failures. This suite uses its existing QA fixtures and is not live provider evidence.

EVIDENCE / AUDIT TRAIL

Sources and reproducibility

Research-site facts: Chaki et al. (2022), Field Crops Research 275, 108344, printed p.3. CIMMYT repository - original research PDF. This walkthrough does not implement the paper's APSIM model or reproduce its yield results.

Regional reference dates: DAE/BAMIS crop-weather calendars. Each output interval retains its original document URL in the export and application.

NASA: NASA POWER data documentation. Actual request URLs and timestamps are retained in the captured evidence.

Weather and places: Open-Meteo weather documentation and GeoNames place search via Open-Meteo. Map attribution remains on the map. Crop photos have separate credits in the application.

Local audit evidence: research/input-audit-2026-10-31/results.json; browser-results/*/scenario.json; walkthrough/evidence.json; walkthrough/ritu-farm-records.json; authentic-provider-records.json; matrix.log. The export hash below identifies the exact definitive run.

Data availability and weather values change over time. This document records a dated run, not a promise that the same live values will appear on a later visit.

Definitive export SHA-256:
0cb4a5983636e8439fe4b1d56ad7f731
32fc6bf68db07fc9d514bfbd4526903c

Captured: 2026-10-31T13:00:13.767Z